EXCELLENCE

CLASS 11 · 2026–27

Physics | Lecture - 05

Basic Maths and Calculus

Last-week formulae, then PW Lecture 04

By — Excellence Programme

8 September 2026

Topics to be covered

01 Last week in 10 minutes — lock-lines only, no new working

02 Last class stopped at slide 55. Today starts from slide 56 — projectile range

03 Finish maximum–minimum, then PW phase / phasor and superposition

04 PW Lecture 04 — angle tool, arithmetic progression and geometric progression

05 Physics uses of G.P. — electric potential and repeating resistance blocks

06 Graph basics — axes, constant lines and distance between two points

Last week — lock-lines only

Last class stopped at slide 55. Today we do not re-teach that board. Say these lines once.

180° = π rad **Arc = R θ** **(θ in radian)**

sin (A + B) = sin A cos B + cos A sin B

cos (A + B) = cos A cos B − sin A sin B

sin θ ≈ θ **tan θ ≈ θ** **cos θ ≈ 1** **(θ in radian, θ ≤ 5°)**

sin 2° is not 2. Convert first: 2 × (π/180) = π/90.

ASTC and the max–min we already locked

1. Sign from ASTC. 2. Function changes at 90° and 270° only. 3. sin ↔ cos.

$(\sin\theta)_{max} = +1$ at $90°$ **$(\sin\theta)_{min} = -1$ at $270°$**

$$y = A\sin(n\theta) \qquad y_{max} = |A| \qquad \text{at } n\theta = 90°$$

Last board, slide 55: $y = 4\sin(3\theta) \rightarrow y_{max} = 4$ at $\theta = 30°$

Do not set $\theta = 90°$. The sine is of 3θ. Divide $90°$ by 3. We start from slide 56 now.

Last class, slide 56 — range of a projectile

Last class, slide 56. If $R = u^2 \sin(2\theta) / g$, with u and g constant, find θ at which R is maximum.

Rmax when sin(2θ) is maximum, so sin(2θ) = 1 2θ = 90° θ = 45°

θ = 45° Rmax = u^2 / g

This is why a level-ground projectile goes farthest at 45°.

Rough work / extra work

Add like sines first

If $y = 3\sin\theta + 4\sin\theta$, find ymax.

$y = 7\sin\theta$ $\quad$ $y_{max} = 7 \times 1 = 7$ $\quad$ at $\theta = 90°$

ymax = 7

Do not open $\sqrt{(3^2+4^2)}$ here. Both terms are sine of the same angle. Add the 3 and the 4.

A constant sitting next to the sine

Find ymax and ymin.

$y = 3 + 2\sin\theta$ $\quad ymax = 3 + 2(1) = 5$ $\quad ymin = 3 + 2(-1) = 1$

$y = 3 + 4\sin\theta$ $\quad ymax = 7$ $\quad ymin = 3 - 4 = -1$

$y = 3\sin\theta - 4$ $\quad ymax = 3 - 4 = -1$ $\quad ymin = -3 - 4 = -7$

$3 + 2\sin\theta \rightarrow 5$ and 1 $\quad 3 + 4\sin\theta \rightarrow 7$ and -1 $\quad 3\sin\theta - 4 \rightarrow -1$ and -7

When sine and cosine are added

sin θ and cos θ do not peak together. The combined amplitude is the hypotenuse of a and b.

$y = a \sin\theta + b \cos\theta$

$$y_{max} = +\sqrt{(a^2 + b^2)} \qquad y_{min} = -\sqrt{(a^2 + b^2)}$$

Do not write ymax = a + b. That would need sin = 1 and cos = 1 at the same instant, which never happens.

$y = 3 \sin \theta + 4 \cos \theta$

If $y = 3 \sin \theta + 4 \cos \theta$, find ymax and ymin.

$y_{max} = \sqrt{(3^2 + 4^2)} = \sqrt{(9 + 16)} = \sqrt{25} = +5$

$y_{min} = -5$

$y_{max} = +5 \qquad y_{min} = -5$

Minimum magnitude of a fraction

If $y = 4 / (3 \sin\theta + 2 \cos\theta)$, find the minimum magnitude of y.

4/10 = 0.4. 4/100 = 0.04. A larger denominator makes a smaller positive fraction.

$y_{min} = 4 / (3 \sin\theta + 2 \cos\theta)_{max} = 4 / \sqrt{(3^2 + 2^2)} = 4 / \sqrt{13}$

minimum magnitude $= 4 / \sqrt{13}$

Rough work / extra work

Five max–min — every line

Find maximum and minimum of each.

	y	ymax	ymin
(a)	$4\sin\theta + 3\cos\theta$	$\sqrt{(16+9)} = 5$	-5
(b)	$2\sin\theta - 3\cos\theta$	$\sqrt{(4+9)} = \sqrt{13}$	$-\sqrt{13}$
(c)	$1 + 5\sin\theta$	$1+5 = 6$	$1-5 = -4$
(d)	$1 - 2\sin\theta$	$1-2(-1) = 3$	$1-2(1) = -1$
(e)	$5\sin\theta - 12\cos\theta$	$\sqrt{(25+144)} = 13$	-13

BREAK

Max–min leftover is done. Next: phase and phasor

Phase is the angle inside the ratio

Phase is the angle sitting inside the trigonometric function. Not the 4 in front.

$\sin \theta$	$\text{phase} = \theta$
$\sin (\theta + \alpha)$	$\text{phase} = \theta + \alpha$
$\sin (\omega t + \alpha + 90°)$	$\text{phase} = \omega t + \alpha + 90°$

Later: AC, SHM, waves, wave optics, EM waves, vectors.

Phase difference — same function first

Angle difference between two of the same trigonometric function is called phase difference.

$y_1 = \sin\theta$ **$\varphi_1 = \theta$**

$y_2 = \sin(\theta + 60°)$ **$\varphi_2 = \theta + 60°$**

$\Delta\varphi = \varphi_2 - \varphi_1 = +60°$ **(y2 leads y1)** **$\Delta\varphi = \varphi_1 - \varphi_2 = -60°$** **(lag)**

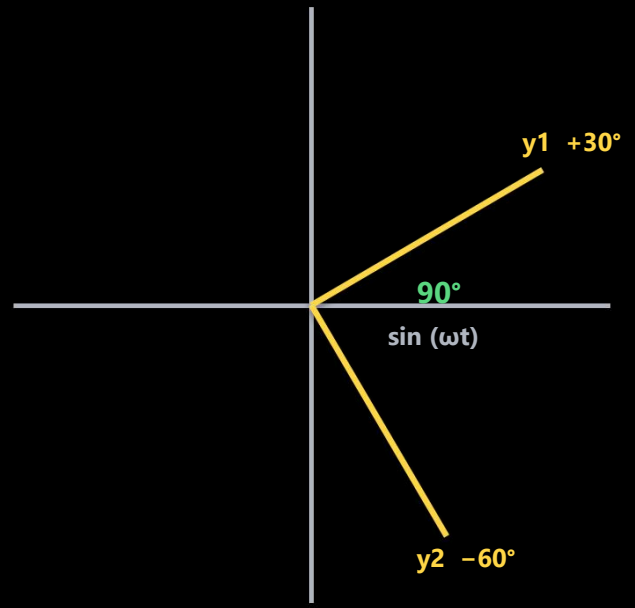
Do not subtract a cosine phase from a sine phase until both are written as the same ratio.

y1 leads y2 by 90°

Find the phase difference. $y1 = \sin(\omega t + 30°)$ $y2 = \sin(\omega t - 60°)$

$\Delta\varphi = \varphi1 - \varphi2 = (\omega t + 30°) - (\omega t - 60°) = 90°$

$\Delta\varphi = 90°$ — y1 is ahead of y2



Rough work / extra work

Convert to the same ratio, or the answer is wrong

Find Δφ. $y1 = 3\cos(\omega t - 90°)$ $y2 = 4\sin(\omega t)$

Wrong: $\Delta\varphi = \omega t - (\omega t - 90°) = 90°$

Right: first write both as sine (or both as cosine)

$\cos(\omega t - 90°) = \sin \omega t$ so $y1 = 3\sin(\omega t)$

$\Delta\varphi = 0$ — they are in phase

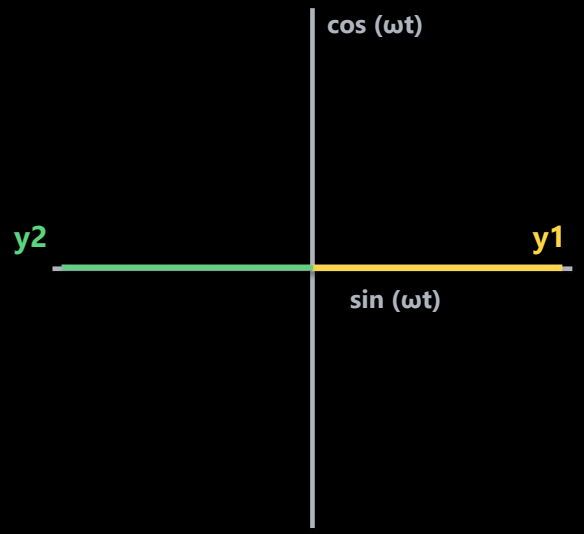
A minus cosine is a sine with a shift. Convert, then subtract.

cos (ωt + π/2) points opposite to sine

Find Δφ. y1 = 3 sin (ωt) y2 = 4 cos (ωt + π/2)

cos (ωt + π/2) = − sin (ωt) so y2 = − 4 sin (ωt)

Δφ = 180° = π rad — opposite on the phasor



Two more phase differences

(1) $x_1 = A\cos(\omega t - \pi/2)$ $\quad y_2 = A\cos(\omega t + \pi/3)$

Same function already. $\Delta\varphi = (\omega t + \pi/3) - (\omega t - \pi/2) = 5\pi/6 = 150°$

(1) $\Delta\varphi = 150°$

(2) $y_1 = 4\sin(\theta + \pi/4)$ $\quad y_2 = 3\sin(\theta - \pi/2)$

$\Delta\varphi = (\theta + \pi/4) - (\theta - \pi/2) = \pi/4 + \pi/2 = 135°$

(2) $\Delta\varphi = 135°$ $\qquad$ Also: $y_1 = 4\sin(\theta+\pi/3)$, $y_2 = 3\sin(\theta+\pi/6) \rightarrow 30°$

AC current and voltage

$I = I_0 \sin(\omega t - 30°)$ $V = V_0 \cos(\omega t)$. Find the phase difference.

$V = V_0 \cos(\omega t) = V_0 \sin(\omega t + 90°)$

$\Delta\varphi = 90° - (-30°) = 120° = 2\pi/3$

$\Delta\varphi = 2\pi/3$ — voltage leads current by 120°

Negative cosine is a shifted sine

$I = -I_0 \cos(\omega t)$ $V = V_0 \sin(\omega t - \pi/6)$. Find $\Delta\varphi$.

1. $2\pi/3$
2. $\pi/3$
3. $\pi/6$
4. $5\pi/6$

$-\cos(\omega t) = \sin(\omega t - \pi/2)$ $\Delta\varphi = (-\pi/6) - (-\pi/2) = \pi/3$

Option 2 $\Delta\varphi = \pi/3$

Rough work / extra work

Person walks toward a 40 m building

A person sees a 40 m building at elevation 30°, then at 60°. Distance moved toward the building?

1. $40/\sqrt{3}$
2. $80/\sqrt{3}$
3. $60/\sqrt{3}$
4. $120/\sqrt{3}$

$x1 = 40 \cot 30° = 40\sqrt{3}$ $\quad x2 = 40 \cot 60° = 40/\sqrt{3}$

$d = x1 - x2 = 40\sqrt{3} - 40/\sqrt{3} = 40(3 - 1)/\sqrt{3} = 80/\sqrt{3}$

Option 2 $\quad d = 80/\sqrt{3}$

$2\,(\sin 15° + \sin 75°)^2$

What is the value of $2\,(\sin 15° + \sin 75°)^2$?

$\sin 75° = \cos 15°$ so $2\,(\sin 15° + \cos 15°)^2$

$= 2\,(\sin^2 15° + \cos^2 15° + 2 \sin 15° \cos 15°) = 2\,(1 + \sin 30°)$

$= 2\,(1 + 1/2) = 3$

$2\,(\sin 15° + \sin 75°)^2 = 3$

If sin θ = 1/2, first quadrant

If sin θ = 1/2, the value of √3 (cos θ + 2 tan θ) is:

1. 7
2. 15/2
3. 7/2
4. none

First quadrant: θ = 30° cos 30° = √3/2 tan 30° = 1/√3

√3 (√3/2 + 2/√3) = 3/2 + 2 = 7/2

Option 3 7/2

If $\sin\theta = \sqrt{5}/3$

If $\sin\theta = \sqrt{5}/3$, find $\tan\theta + 4\sec\theta$. Take θ acute.

$\cos\theta = \sqrt{(1 - 5/9)} = \sqrt{(4/9)} = 2/3$ $\tan\theta = (\sqrt{5}/3) / (2/3) = \sqrt{5}/2$

$\sec\theta = 3/2$ $\tan\theta + 4\sec\theta = \sqrt{5}/2 + 4 \times (3/2) = \sqrt{5}/2 + 6$

$\tan\theta + 4\sec\theta = 6 + \sqrt{5}/2$

$\tan^{-1}(1)$ — principal value

The value of $\tan^{-1}(1)$ is

1. $\pi/4$
2. $5\pi/4$
3. both (1) and (2)
4. none of these

Wrong: $\tan(5\pi/4) = 1$, so both

Right: $\tan^{-1}$ returns only the principal value, in $(-\pi/2, \pi/2)$

Option 1 $\pi/4$

$5\pi/4$ is an angle whose tangent is 1. It is not the inverse-tangent of 1.

Rough work / extra work

Seven phase rows — answers from the PW board

Complete the phase-difference table from last class.

Equation 1	Equation 2	phase difference
$I_0 \sin(\theta + \pi/3)$	$I_0 \sin(\theta - \pi/6)$	90°
$I_0 \sin(\theta + \pi/3)$	$I_0 \cos(\theta + \pi/3)$	90°
$4 \sin(\omega t)$	$4 \cos(\omega t - \pi/3)$	30°
$3 \sin(\omega t + \pi/2)$	$5 \cos(\omega t)$	0°
$4 \cos(\omega t - \pi/2)$	$4 \sin(\omega t + \pi/2)$	90°
$3 \sin(\omega t + \pi/3)$	$3 \cos(\omega t)$	30°
$4 \sin(\theta - 60°)$	$4 \cos(\theta - 30°)$	120°

Amplitude does not decide phase. Compare only the angles after both functions are the same.

Superposition — add two oscillations as phasors

For the same angular frequency, add the sine part and cosine part like perpendicular sides.

$$x = a\sin(\omega t) + b\cos(\omega t) = R\sin(\omega t + \varphi)$$

$$R = \sqrt{(a^2 + b^2)} \qquad \tan\varphi = b/a$$

R is the new amplitude. φ tells how far the resultant phasor is above the sine axis.

This shortcut needs the same ω in both terms. Do not combine sin(2ωt) with cos(ωt).

PW example: $4 \sin(\omega t) + 3 \cos(\omega t)$

Two oscillations are $x_1 = 4 \sin(\omega t)$ and $x_2 = 3 \cos(\omega t)$. Find their superposition.

$x = x_1 + x_2 = 4 \sin(\omega t) + 3 \cos(\omega t)$

$R = \sqrt{(4^2 + 3^2)} = 5 \qquad \tan \varphi = 3/4 \qquad \varphi \approx 37°$

$x = 5 \sin(\omega t + 37°)$

PW example: equal sine and cosine parts

Find the resultant: $y = 4\sin(\omega t) + 4\cos(\omega t)$.

$R = \sqrt{(4^2 + 4^2)} = \sqrt{32} = 4\sqrt{2}$

$\tan\varphi = 4/4 = 1$ so $\varphi = 45°$

$y = 4\sqrt{2}\sin(\omega t + 45°)$

Angle tool — incline and parallel lines

A line perpendicular to an incline is called the normal. If the incline is θ above horizontal:

angle between vertical and normal = θ

For two parallel horizontal lines cut by one slant line, alternate interior angles are equal.

alternate angles: α = α **triangle angles: A + B + C = 180°**

Do not use the angle with the incline when the force component is measured from the normal.

Rough work / extra work

PW angle check — find α, β and γ

Two strings make 30° and 60° with a horizontal ceiling. Find α, β and γ at the lower point.

alternate angles → α = 30° γ = 60°

α + β + γ = 180° β = 180° − 30° − 60° = 90°

α = 30° β = 90° γ = 60°

A.P. — equal difference every time

Arithmetic progression is a sequence in which the difference between consecutive terms is constant.

a, a+d, a+2d, a+3d, ...

nth term: $T_n = a + (n-1)d$

sum: $S_n = n/2\ [2a + (n-1)d]$

a = first term. d = next term − previous term. n = position of the term.

PW A.P.: 2, 5, 8, 11, 14, ...

Find the first term, common difference, 10th term and sum of first 10 terms.

$a = 2 \qquad d = 11 - 8 = 3$

$T_{10} = a + 9d = 2 + 9(3) = 29$

$S_{10} = 10/2\ [2 + 29] = 5(31) = 155$

$T_{10} = 29 \qquad S_{10} = 155$

PW A.P.: 7, 11, 15, 19, ...

Find the 10th term and the sum of the first 10 terms.

$a = 7 \qquad d = 19 - 15 = 4$

$T_{10} = 7 + 9(4) = 43$

$S_{10} = 10/2\ (7 + 43) = 5(50) = 250$

10th term = 43 sum = 250

Which term of 27, 24, 21, ... is zero?

Which term of the A.P. 27, 24, 21, ... is zero?

$a = 27 \qquad d = 24 - 27 = -3$

$0 = a + (n-1)d = 27 + (n-1)(-3)$

$0 = 30 - 3n \qquad 3n = 30 \qquad n = 10$

The 10th term is zero.

Rough work / extra work

PW A.P.: 28, 22, x, y, 4

28, 22, x, y, 4 is an A.P. Find x and y.

$a = 28 \qquad d = 22 - 28 = -6$

$x = T_3 = 28 + 2(-6) = 16$

$y = T_4 = 28 + 3(-6) = 10$

$x = 16 \qquad y = 10$

Natural numbers are an A.P.

1, 2, 3, ..., n is an A.P. with first term 1, last term n and n terms.

$$1 + 2 + 3 + \ldots + n = n(n+1)/2$$

Quick check: n = 2 gives 1 + 2 = 3. The formula gives 2×3/2 = 3.

PW physics use — tuning-fork frequencies

25 tuning forks are in increasing A.P. order. Difference is 10 Hz. The 25th is 400 Hz. Find the first.

$T_{25} = a + 24d$ $\quad\quad$ $400 = a + 24(10)$

$a = 400 - 240 = 160$ Hz

Frequency of the first tuning fork = 160 Hz

G.P. — equal ratio every time

Geometric progression is a sequence in which the ratio of consecutive terms is constant.

$a,\ ar,\ ar^2,\ ar^3,\ \ldots$

nth term: $T_n = ar^{n-1}$

infinite sum: $S\infty = a/(1-r)$, only if $|r| < 1$

r = any term ÷ previous term. If that ratio changes, the sequence is not a G.P.

Is it a G.P.? Check the ratio

Decide which sequences are G.P.

sequence	ratio check	answer
2, 4, 8, 12, 24, 48	4/2=2, but 12/8≠2	not G.P.
2, 4, 8, 16, 32, 64	each ratio = 2	G.P., r=2
1, 1/2, 1/4, 1/8, …	each ratio = 1/2	G.P., r=1/2
1, −1/2, 1/4, −1/8, …	each ratio = −1/2	G.P., r=−1/2

Rough work / extra work

Infinite G.P. sums — only when |r| < 1

Find each infinite sum.

(1) $1 + 1/2 + 1/4 + \ldots = 1/(1-1/2) = 2$

(2) $1 - 1/2 + 1/4 - \ldots = 1/[1-(-1/2)] = 2/3$

(3) $1 + 1/3 + 1/9 + \ldots = 1/(1-1/3) = 3/2$

$2 + 4 + 8 + \ldots$ has $r = 2$. It has no finite infinite sum because $|r|$ is not below 1.

PW physics use — potential from alternating charges

Charges +q, −q, +q, … lie at x = 1, 2, 4, 8, … . Find electric potential at the origin.

$V = kq/1 - kq/2 + kq/4 - kq/8 + \dots$

$V = kq\,[1 - 1/2 + 1/4 - \dots]$ $\quad a=1,\ r=-1/2$

$V = kq \,/\, [1 - (-1/2)] = 2kq/3$

Potential at origin = $2kq/3$

PW physics use — repeating resistance blocks

Equivalent series resistance becomes R + R/2 + R/4 + R/8 + … . Find R_AB.

R_AB = R [1 + 1/2 + 1/4 + 1/8 + …]

a = 1, r = 1/2 sum = 1/(1−1/2) = 2

R_AB = 2R

Graph — relation between two physical quantities

A graph is a diagram in which a line or curve shows the relation between two physical quantities.

horizontal line = x-axis **vertical line = y-axis** **meeting point = origin (0,0)**

Usually x is the quantity we choose or change. y is the quantity that responds.

Always write: quantity + unit on each axis.

Constant graphs — horizontal and vertical

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y = 2 → horizontal line through y = 2

x can change, but y stays 2. So the graph moves left-right at the same height.

x = 3 → vertical line through x = 3

y can change, but x stays 3. So the graph moves up-down at the same x-position.

y = constant is horizontal. x = constant is vertical. Students commonly swap them.

Rough work / extra work

Distance between two points on a graph

For $P_1(x_1,y_1)$ and $P_2(x_2,y_2)$, the horizontal change is x_2-x_1 and vertical change is y_2-y_1.

$$P_1P_2 = \sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2]}$$

It is simply Pythagoras on the small right triangle made by the two changes.

PW Lecture 04 homework — three questions

(1) Charges q lie at x = 1, 2, 4, 8, … . Find the force on charge q_0 at the origin.

(2) Five resistances are in series. Each is half the previous one. First is 10 Ω. Find the fifth resistance.

(3) Charges +q, −q, +q, … lie at x = 1, 2, 4, 8, … . Find potential at the origin.

Hint only: identify a, identify r, then choose T_n or S_∞.

Do not solve these on this slide. They are tonight's work.

After class

Daily test before the next lecture.
Write A.P. nth-term and sum formula five times.
Write G.P. nth-term and infinite-sum formula five times.

Tonight: the three PW Lecture 04 homework questions.

Max–min — the two boxes

$y = A \sin(n\theta)$ $\quad y_{max} = |A|$ $\quad$ at $n\theta = 90°$

$$y = a \sin\theta + b \cos\theta \qquad y_{max} = \sqrt{(a^2 + b^2)}$$

$3 \sin\theta + 4 \cos\theta \rightarrow \pm 5$ $\quad 4 / (3 \sin\theta + 2 \cos\theta) \rightarrow 4/\sqrt{13}$

3 sin θ + 4 sin θ is 7 sin θ, not √(9+16). Add like ratios first.

Phase — the one lock-line

Phase is the angle inside the ratio. Phase difference needs the same trigonometric function.

$\Delta\varphi = \varphi_2 - \varphi_1$ **after both are sine, or both are cosine**

$3\cos(\omega t - 90°)$ and $4\sin(\omega t)$ → $\Delta\varphi = 0$

Never subtract a cosine-phase from a sine-phase. Convert, then subtract.

Daily test before the next lecture. Three PW A.P./G.P. questions tonight.

Rough work / extra work

A.P., G.P. and graph — lock-lines

A.P.: $T_n = a + (n-1)d$ $S_n = n/2\,[2a + (n-1)d]$

G.P.: $T_n = ar^{n-1}$ $S\infty = a/(1-r)$, only if $|r| < 1$

$1 + 2 + \ldots + n = n(n+1)/2$

distance $= \sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2]}$

y = constant → horizontal x = constant → vertical